

Collaboration-Based Learning Environments using Augmented Reality

by
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Submitted to the Department of Creative Technologies
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ABSTRACT

The developments of the “kinetic theory” of gases made within the last ten years have enabled it to account satisfactorily for many of the laws of gases. The mathematical deductions of Clausius, Maxwell and others, based upon the hypothesis of a gas composed of molecules acting upon each other at impact like perfectly elastic spheres, have furnished expressions for the laws of its elasticity, viscosity, conductivity for heat, diffusive power and other properties. For some of these laws we have experimental data of value in testing the validity of these deductions and assumptions. Next to the elasticity, perhaps the phenomena of the viscosity of gases are best adapted to investigation.¹

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¹Text from Holman (1876): doi:[10.2307/25138434](https://doi.org/10.2307/25138434).

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Biographical Sketch

Silas Whitcomb Holman was born in Harvard, Massachusetts on January 20, 1856. He received his S.B. degree in Physics from MIT in 1876, and then joined the MIT Department of Physics as an Assistant. He became Instructor in Physics in 1880, Assistant Professor in 1882, Associate Professor in 1885, and Full Professor in 1893. Throughout this period, he struggled with increasingly severe rheumatoid arthritis. At length, he was defeated, becoming Professor Emeritus in 1897 and dying on April 1, 1900.

Holman's light burned brilliantly before his tragic and untimely death. He published extensively in thermal physics, and authored textbooks on precision measurement, fundamental mechanics, and other subjects. He established the original Heat Measurements Laboratory. Holman was a much admired teacher among both his students and his colleagues. The reports of his department and of the Institute itself refer to him frequently in the 1880's and 1890's, in tones that gradually shift from the greatest respect to the deepest sympathy.

Holman was a student of Professor Edward C. Pickering, then head of the Physics department. Holman himself became second in command of Physics, under Professor Charles R. Cross, some years later. Among Holman's students, several went on to distinguish themselves, including: the astronomer George E. Hale ('90) who organized the Yerkes and Mt. Wilson observatories and who designed the 200 inch telescope on Mt. Palomar; Charles G. Abbot ('94), also an astrophysicist and later Secretary of the Smithsonian Institution; and George K. Burgess ('96), later Director of the Bureau of Standards.

Contents

<i>List of Figures</i>	6
<i>List of Tables</i>	7
List of Acronyms	8
1 Introduction	9
1.1 Context	12
1.2 Research Structure	15
1.2.1 Research Questions	16
1.2.2 Scope and Limitations	16
1.3 Significance and Contributions	16
1.4 Document Structure	16
References	17

List of Figures

1.1	Virtual Classroom in Cyberpunk: Edgerunners	10
1.2	Augmented Reality (AR) in the Milgram-Weiser Spectrum (Schmalstieg, 2016).	13
1.3	Classification of Computer Software for Cooperative Work (CSCW) Systems by Rodden (1991).	14

List of Tables

List of Acronyms

VR Virtual Reality

AR Augmented Reality

TEL Technology Enhanced Learning

MR Mixed Reality

CSCW Computer Software for Cooperative Work

Chapter 1

Introduction

In the Netflix animated series *Cyberpunk: Edgerunners* (Imaishi et al., 2022), the main character attends courses at a prestigious academy that uses advanced technology to boost education and performance. The students gather in a room with Virtual Reality (VR) headsets and connect to the lesson, which is taught by a holographic (and probably powered by artificial intelligence) tutor. The episode does not show the lesson itself (the entire system fails catastrophically when attacked by malware), but the brief scene highlights the vision many people have of a futuristic classroom, one guided by advanced technology and automatization.

Forsler et al. (2024) propose the term “post-digital classroom” to identify the characteristics and trends in education present in a world beyond the adoption of digital technologies. The purpose of this classroom is not anymore to introduce new technologies to students, but to implement them as an essential factor of the learning process. The post-digital classroom is interconnected, social and global. In this scenario, learning goes beyond the physical classroom because it is based on creating relationships between concepts and developing valuable skills rather than acquiring knowledge. It is also a classroom where information is detached from the physical learning institute, and access to facts and sources is, ideally, immediate, ubiquitous and democratic.

The classroom has also become hybrid. There is little distinction between network-based lessons and the face-to-face spaces associated with schools and universities (Goodyear et al., 2004). Hybrid and blended approaches blur the distinctions between learning in an online setting and learning in a classroom. In both instances, information is at hand in the network, students can easily connect to peers and mentors, distance and time constraints are less meaningful and knowledge is presented in a multimedia format that needs a technological backbone to be created and shared (Rivoltella, 2008; Shank, 2005).

This thesis explores the use of technology in such post-digital classrooms, specifically, AR within the context of mobile computing. Moreover, this research will analyse the creation of effective learning spaces, those places where students can build, personalise and control all the activities they engage with (Gourlay & Oliver, 2016). Learning spaces are interesting tools at the disposal of students because these are spaces that can exist outside the norm of a typical classroom, they are more fluid, less structured and can easily connect to other similar spaces (Nørgård & Hilli, 2022). The proposal of this research is that AR can be used to boost the way students build and interact with their own learning spaces and can also



Figure 1.1: Virtual Classroom in Cyberpunk: Edgerunners

offer educators another option to be integrated in the construction of a lesson that capitalises the characteristics of the aforementioned post-digital classrooms.

Within the field of Technology Enhanced Learning (TEL), AR and related technologies were chosen as a focus of interest because of their intrinsic relationship with space. The initial example scene of Cyberpunk: Edgerunners serves to illustrate the collective idea of what could be achieved in the future with technologies like VR and AR: the possibility of creating an entire new space, suited for the needs of the classroom or the students. It is a space that is shared, in which information and interactions flow between users and that can be controlled at will by students and teachers. In reality, the technology is not quite there yet, especially in terms of fidelity and immersion (Checa & Bustillo, 2020; Hamad & Jia, 2022). What can be explored right now is the creation of interesting synthetic environments that facilitate or allow experiences that are not possible in the common classroom. This process is limited more by the creativity and expertise of the developer rather than by the technology, and therefore can be explored easier and in more detail.

AR, compared to VR, has a more direct relationship with the *real*, physical space and with the context of the user. Instead of an isolated immersion, AR creates a connection with the environment based on image detection, data and visualizations. This natural expression of the technology is where the value of AR can be explored in relation to the creation of innovative learning spaces. The technology not only has interesting capabilities for visualization and interaction, it also allows an easier implementation of shared, social components that can be used to build activities based on cooperative simulations, sharing knowledge and the creation of communities, all crucial elements for the construction of effective learning spaces (Bligh & Crook, 2017).

The research is positioned then in the conjunction of these three fields: Learning spaces, TEL and AR. The guiding goal is to understand how students are using technology to learn,

and how [AR](#) can be used to boost that process. [AR](#) shines at creating immersive experiences that can be shared with others and that consider the physical context of the user. I want to use this collaborative space as an educational tool that takes advantage of the inherent social aspects of learning, building and sharing knowledge.

No modern classroom is free of technology, the simple act of taking notes on a paper already implies a set of technological developments needed to put knowledge into written form. Networking and information technologies are just another set of tools at the disposition of any student. A personal learning space can incorporate different software tools, many of which are now ubiquitous and almost indispensable in any learning activity: text processors, multimedia editing tools, web browsers and document readers, plus any specialised tool that the learning subject requires. But that learning space is also built around more elements, some of them equally indispensable for some students. People need a space to work, access to different amenities, access to textbooks, information and teachers. Some may need a quiet space to concentrate, or, on the contrary, may need access to music, to snacks or to a separate place to periodically take a break. The construction of a learning space involves many different elements, some more technological than others, and its effectiveness is based more on the integration of those elements than anything else (Lu & Churchill, 2013; Reinius et al., 2021). This research aims at identifying how to properly integrate [AR](#) in the construction of a learning space, which is already composed by a complex web of many other objects, tools and circumstances.

As a researcher. I have explored in the past the way different technologies can be used as part of those learning spaces, either with tools for online learning scenarios or using [VR](#) to create a digital space to explore multimedia content (Gil Parga, 2015). This current project can be seen as a way to expand on the work done before, identifying ways to adapt a different technology, to apply lessons learned and to explore new avenues of research. In particular, it is an opportunity to expand from personalized learning environments into social learning spaces, and explore the role of the community in teaching and learning.

It was clear in the work done previously and through some notable literature like Richards (2023) and Long and Ehrmann (2005) that the construction of learning spaces is always mediated by the community around us, even when creating a space as personal as possible. Even nowadays that we live in a post-pandemic world where we seek remote, personalised and secure ways of learning, it has become clear that the social aspect of learning is a crucial element in the equation (Baber, 2021; De Felice et al., 2023). The aim of this research is to delve deeper into the social elements that influence the construction of a learning space, how the students negotiate the process of learning with others and the benefits of collaboration, as well as the challenges it conveys.

This is also a technological enquiry. Having explored [VR](#) in the past, it was clear that this family of technologies offer interesting advantages by using complex visuals, new forms of presenting information and more natural interactions suited for the exploration and experimentation with 3D spaces. These technologies also present curious challenges, especially in the usability front. For instance, the public has not yet developed a strong relation or consistent literacy with the technology (Özgen et al., 2021), which creates accessibility problems. And those technologies that leverage common and accessible platforms, like mobile devices, risk carrying with them problems associated with the excessive and pervasive presence of such technologies (Forsler & Guyard, 2025; Petrucco, 2021).

The following introductory chapter will expand this brief exploration of the context and motivation for the research proposed, outlining the main research goal and the reasoning behind the formulation of the selected research questions. There will be a brief exposition of the methodology followed to build possible answers, the significance of this research and why the answers to the questions proposed can provide important insights about the way we use technology to learn and how we design technological tools for the classroom. It will be followed by a summary of the results obtained and the contributions achieved, as well as the scope and limitations of the work done. The chapter will conclude with a description of the structure of this thesis document.

1.1 Context

AR technologies are positioned somewhere in the middle of the reality-simulation spectrum proposed by Milgram et al. (1995), a model for display technologies used to show digital or physical data in virtual or synthetic realities. The main characteristic of AR in this spectrum is the incorporation of the real world as part of the experience, making use of different image recognition and composition techniques to identify features in the physical space and overlay them with digital information.

Schmalstieg (2016) adds another layer of analysis to the description of AR-like technologies by using the ubiquitous computing spectrum proposed by Weiser (1991). The model positions different implementations of AR depending on the use of more static or more mobile deployment platforms. In one end of the spectrum are applications that can only be used in singular static terminals, while the other side contemplate platform-independent mobile technologies and ubiquitous systems. Figure 1.2 shows examples of this distribution as proposed by the authors.

In this bi-directional spectrum we can find mobile AR as a family of technologies that offer an in-between mixture of virtual and physical elements in which location, place and context are additional inputs used for interaction with the synthetic reality. Any technology in the spectrum will offer a variety of characteristics that align with one or more descriptions at a time. Mixed Reality (MR), for example, are seen as an extension of both VR and AR, offering seamless interaction and connectivity between the real and the simulated world. It is a blurry line that often must be analysed in a case-by-case approach. No matter what label is used to describe a technology, some agreed upon characteristics must be present for it to be categorized as AR, as stated by Azuma (1997):

- A combination of real space and digital information.
- Interaction with the simulation in real time.
- The simulation is done in a 3D space.

With these descriptors it is possible to position the used of the technology in different fields, independent of the labels used to identify it. Based on the themes introduced in the opening statement of the chapter, the goal is to analyse AR as CSCW. For this task, a useful analytical tool is the classification of CSCW technologies proposed by Rodden (1991)

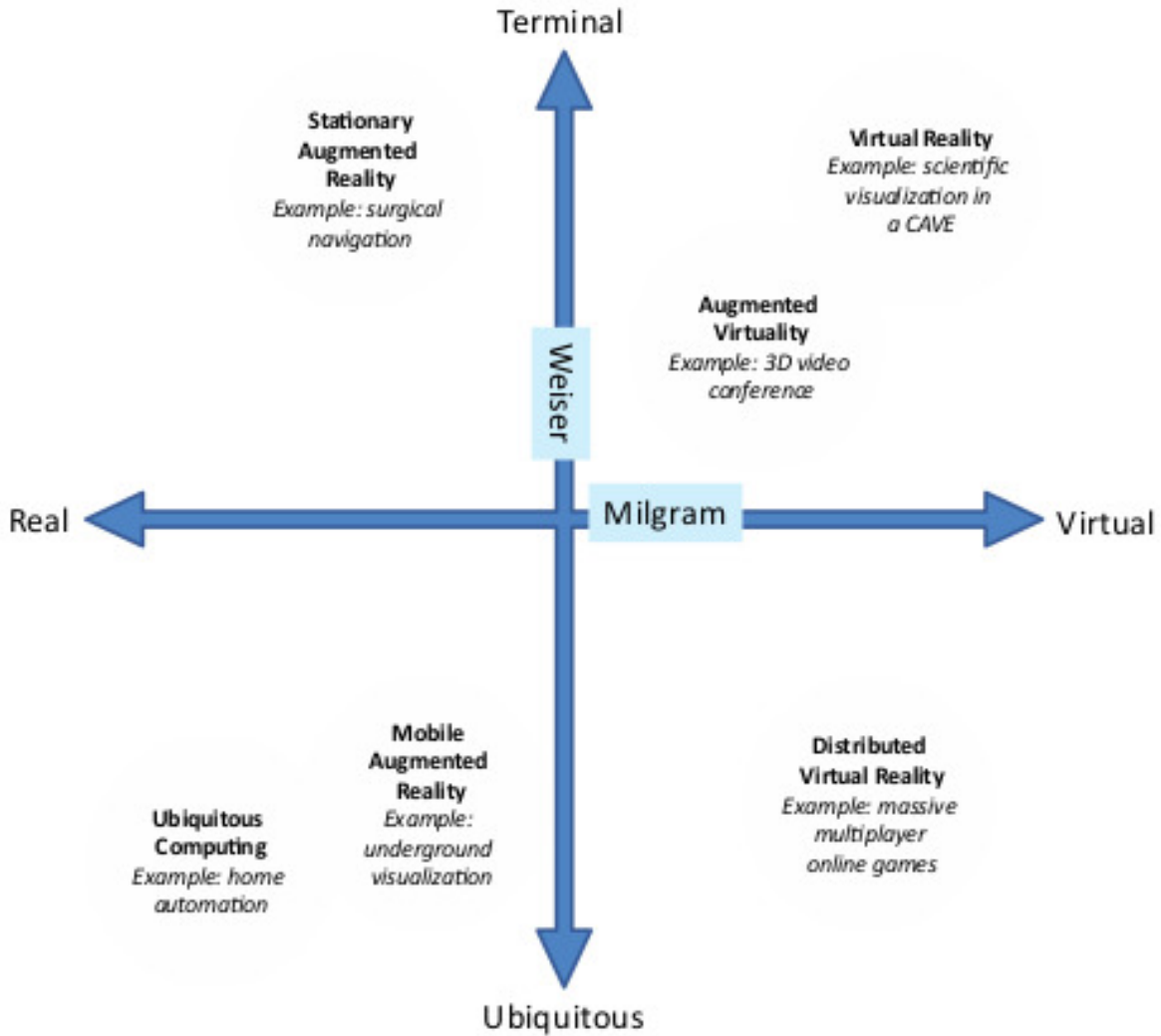


Figure 1.2: AR in the Milgram-Weiser Spectrum (Schmalstieg, 2016).

in terms of the point in time and space where the collaboration is taking place. Figure 1.3 shows examples of different systems that can be found in each quadrant of the classification, which indicate whether the activity is synchronous or asynchronous, remote or co-located.

Complementarily, Kiyokawa et al. (2002) offer two important elements present in a collaborative activity: communication and tasks spaces. In the communication space, people exchange information and can sustain formal and informal conversations. In the task space, the actual cooperative work is done. Those spaces can be unified or separated, and they can be physical, virtual or in a mixed state. Independent of the nature of each individual space, both need to be present for collaboration to succeed.

Based on these classification models, it is possible to define the main field of interest of this research as the use of AR in a mixed task space for face-to-face collaborative learning, which implies synchronous activities in a co-located space. In this setting it should be possible to use the capabilities of AR for visualization and interaction with the physical space to

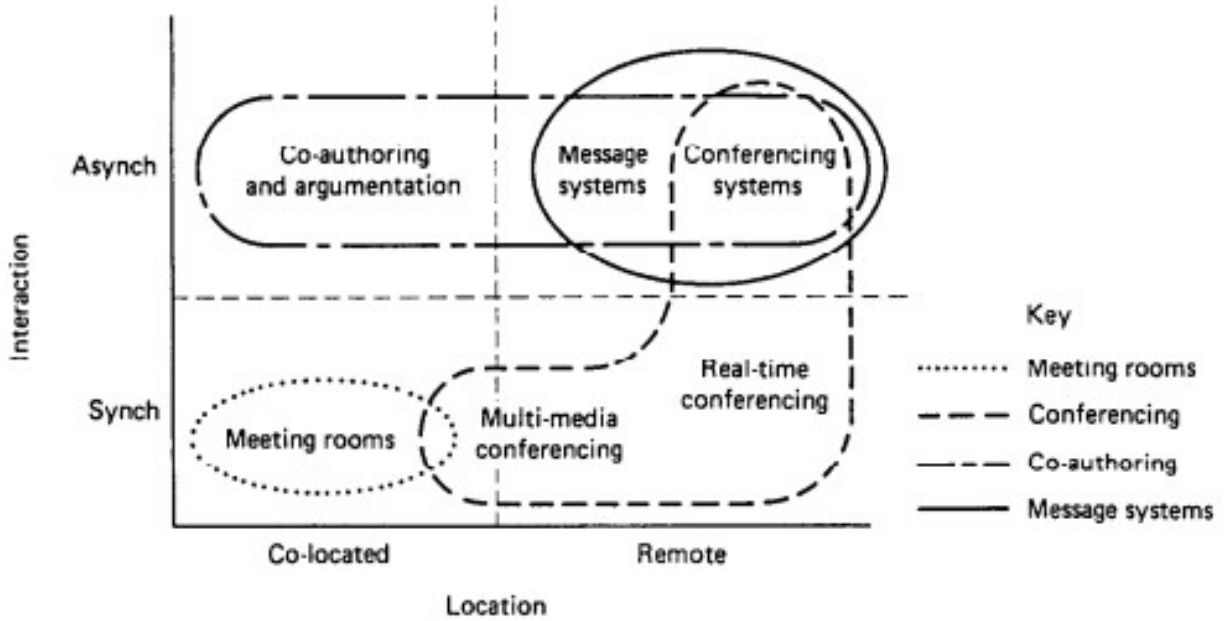


Figure 1.3: Classification of CSCW Systems by Rodden (1991).

augment the collaboration process in terms of:

- Activities related to the communication space, the flow of information and the creation of knowledge in the group.
- The use of the physical space as an asset in the collaborative process.
- Management of the collaborative process itself and the structures that can be created to facilitate effective teamwork and cooperation.
- Guidelines to acquire and refine skills related to teamwork and collaboration.

The main proposal is that the unique characteristics that AR provides in the field of MR are well suited to experiences in co-located collaboration, and that a learning design that is using a collaborative approach can benefit from an AR tool mediating in the communication and tasks spaces of the activity.

The interest in collaborative learning derives from the analysis of the use of AR in education and the tendencies identified in the available literature. It was possible to note that AR is often used as a tool to explore concepts and practices in hands-on approaches that apply previous knowledge in different situations and contexts. Unfortunately, the technology is seldom used in cooperative scenarios, despite the tendency to use mobile platforms for deployment (Gil Parga et al., 2023).

Collaborative learning is a subject rich in discussion. The conversation can focus on the benefits of groupwork in the classroom like promoting different points of view, facilitating complex understanding of a topic and creating the space for an integral development of the student (Smith et al., 1992; Van Wyk & Haffejee, 2017). On the other hand, the challenges associated with collaborative work are also amply explored (Baloche & Brody, 2017; Buchs

et al., 2017; Popov et al., 2012), and there is plenty discussion around issues related to classroom management, inclusivity, effective assessment and good practices.

Many studies have identified and compiled the advantages of using AR and MR as learning tools. For instance, Penn and Ramnarain (2023) offer an analysis of the educational foundations used to design learning interventions. They conclude that most projects converged into constructivist foundations and embodiment theory, opting to create spaces for students to experience and explore the learning subject, with an emphasis on immersion, practice and hands-on interactions. Despite these commonalities, the authors also remarked that most reports do not mention explicitly or do not rationalize the theoretical background of the intervention. They also note that a lot of researchers still hold the idea that introducing a new technology in the classroom is enough to reap its benefits, without caring to build or follow a proper integration plan.

This issue acquires even more relevance now that the mediation of different technologies in the collaboration process has grown due to the Covid-19 pandemic, which forced a quick and often poorly planned implementation of remote learning (Drueke et al., 2021; Nuere et al., 2021). Over time, this prompted the re-design and re-thinking of how to use technology to support the sudden disruption in learning methodologies and approaches (Adi Badiozaman et al., 2020; Fischer et al., 2020). The forced isolation also highlighted the importance of social spaces in the constructions of learning environments, and the need to provide those social interactions even in online spaces (McInnerney & Roberts, 2004; Walas-Trębacz et al., 2025).

The post-pandemic classroom is dealing now with the need to balance the advantages of online spaces with the problems associated with socialization solely mediated by technology, which tends to be asynchronous and impersonal. There is value in the physical space for the learning process (Paechter & Maier, 2010; Petchamé et al., 2021; Thai et al., 2020), and a technology that is able to connect the physical and the digital is worthy of a deeper exploration to identify proper approaches and successful use cases.

The idea is not to oversimplify the presence of digital technologies in the classroom, nor follow a misguided tendency of selling all the benefits of a technology without acknowledging its problems. If the proposal is to identify the effects of a collaborative learning design mediated by AR, it is important to also identify all the problems that derive from this approach, either because of the issues that naturally come from teamworking scenarios in the classroom or the negative behaviours that the excessive use of digital technologies promotes, especially in the specific environment of mobile platforms.

With that context just detailed, it is possible to summarize the goal of this research as: The construction of an AR tool that serves as aid in a collaborative learning setting by helping students to communicate and develop teamworking skills. This can be achieved by means of interactivity, visualization, immersion and augmentation of the physical space.

1.2 Research Structure

The goal of the research can be justified by an abundance of literature showing the elements of AR used in the past to offer distinctive value for learning, and the open opportunities to explore and understand better the development of multi-user AR experiences for collaboration.

The nature of the proposal implies that the research has two main fronts:

- In the *technological* front, the goal is to build a software solution that considers the development challenges of an [AR](#) tool in a multi-user and multi-platform environment, and the specific requirements derived from the learning goals proposed in the educational front.
- In the *educational* front, the goal is to identify, describe and develop a case study based on collaborative learning, and to analyse the behaviours that students develop when learning in a collaborative environment mediated by technology, comparing the use of [AR](#) with other tools or approaches.

1.2.1 Research Questions

The guiding research questions of this project are aimed at obtaining information of what constitutes an effective collaborative learning design using [AR](#). The answers can be stated in terms of learning goals, interaction design and integration plans with the classroom. The questions also seek information about how [AR](#) is creating an effect in the behaviour of the students while using the tool for learning. These behaviours can be related to communication, socialization, planning and engagement with the learning material. The main research question will be stated as follows:

MQ: How does the behaviour of a group of students changes when using an [AR](#) tool designed to aid in the achievement of a learning goal framed in a collaborative learning scenario?

To guide the implementation of the tool through a strong design, a lot of information can be extracted from current literature and from previous projects using similar approaches. Information gathered from real and incremental usage off the system also offers valuable insights in terms of usability, viability and end-user perception. With this in mind, an iterative development process is proposed to facilitate the exploration of ideas regarding interaction design and a proper fit to the learning scenario where the tool is going to be used.

1.2.2 Scope and Limitations

1.3 Significance and Contributions

1.4 Document Structure

References

- Adi Badiozaman, I. F., Leong, H. J., & Wong, W. (2020). Embracing educational disruption: A case study in making the shift to a remote learning environment. *Journal of Applied Research in Higher Education*, 14(1), 1–15.
<https://doi.org/10.1108/JARHE-08-2020-0256>
- Azuma, R. T. (1997). A Survey of Augmented Reality. *Presence: Teleoperators and Virtual Environments*, 6(4), 355–385. <https://doi.org/10.1162/pres.1997.6.4.355>
- Baber, H. (2021). Social interaction and effectiveness of the online learning – A moderating role of maintaining social distance during the pandemic COVID-19. *Asian Education and Development Studies*, 11(1), 159–171.
<https://doi.org/10.1108/AEDS-09-2020-0209>
- Baloche, L., & Brody, C. M. (2017). Cooperative learning: Exploring challenges, crafting innovations. *Journal of Education for Teaching*, 43(3), 274–283.
<https://doi.org/10.1080/02607476.2017.1319513>
- Bligh, B., & Crook, C. (2017). Learning Spaces. In E. Duval, M. Sharples, & R. Sutherland (Eds.), *Technology Enhanced Learning: Research Themes* (pp. 69–87). Springer International Publishing. https://doi.org/10.1007/978-3-319-02600-8_7
- Buchs, C., Filippou, D., Pulfrey, C., & Volpé, Y. (2017). Challenges for cooperative learning implementation: Reports from elementary school teachers. *Journal of Education for Teaching*, 43(3), 296–306. <https://doi.org/10.1080/02607476.2017.1321673>
- Checa, D., & Bustillo, A. (2020). Advantages and limits of virtual reality in learning processes: Briviesca in the fifteenth century. *Virtual Reality*, 24(1), 151–161.
<https://doi.org/10.1007/s10055-019-00389-7>
- De Felice, S., Hamilton, A. F. D. C., Ponari, M., & Vigliocco, G. (2023). Learning from others is good, with others is better: The role of social interaction in human acquisition of new knowledge. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 378(1870), 20210357. <https://doi.org/10.1098/rstb.2021.0357>
- Drueke, B., Mainz, V., Lemos, M., Wirtz, M. A., & Boecker, M. (2021). An Evaluation of Forced Distance Learning and Teaching Under Pandemic Conditions Using the Technology Acceptance Model. *Frontiers in Psychology*, 12.
<https://doi.org/10.3389/fpsyg.2021.701347>
- Fischer, G., Lundin, J., & Lindberg, J. O. (2020). Rethinking and reinventing learning, education and collaboration in the digital age—from creating technologies to transforming cultures. *The International Journal of Information and Learning Technology*, 37(5), 241–252. <https://doi.org/10.1108/IJILT-04-2020-0051>

- Forsler, I., Bardone, E., & Forsman, M. (2024). The Future Postdigital Classroom. *Postdigital Science and Education*. <https://doi.org/10.1007/s42438-024-00488-y>
- Forsler, I., & Guyard, C. (2025). Screens, teens and their brains. Discourses about digital media, learning and cognitive development in popular science neuroeducation. *Learning, Media and Technology*, 50(2), 191–204. <https://doi.org/10.1080/17439884.2023.2230893>
- Gil Parga, S. (2015). *Ambientes virtuales de aprendizaje aplicados* [Doctoral dissertation, Universidad de los Andes]. Retrieved July 7, 2025, from <https://repositorio.uniandes.edu.co/entities/publication/c5cf341b-1dfe-43bd-ac7c-61294516c8c2>
- Gil Parga, S., Singh, U., Gutierrez, J., & Marks, S. (2023). Pedagogical design in education using augmented reality: A systematic review. *Interactive Learning Environments*, 1–18. <https://doi.org/10.1080/10494820.2023.2195445>
- Goodyear, P., Banks, S., Hodgson, V., & McConnell, D. (2004). Research on networked learning: An overview. In P. Dillenbourg, M. Baker, C. Bereiter, Y. Engeström, G. Fischer, H. Ulrich Hoppe, T. Koschmann, N. Miyake, C. O'Malley, R. Pea, C. Pontecorovo, J. Roschelle, D. Suthers, P. Goodyear, S. Banks, V. Hodgson, & D. McConnell (Eds.), *Advances in Research on Networked Learning* (pp. 1–9). Springer Netherlands. https://doi.org/10.1007/1-4020-7909-5_1
- Gourlay, L., & Oliver, M. (2016). Students' Physical and Digital Sites of Study: Making, Marking, and Breaking Boundaries. In *Place-Based Spaces for Networked Learning* (pp. 73–86). Routledge.
- Hamad, A., & Jia, B. (2022). How Virtual Reality Technology Has Changed Our Lives: An Overview of the Current and Potential Applications and Limitations. *International Journal of Environmental Research and Public Health*, 19(18). <https://doi.org/10.3390/ijerph191811278>
- Imaishi, H., Kaneko, Y., Pondsmith, M., Sztybor, B., & Ôtsuka, M. (2022, September). Cyberpunk: Edgerunners (episode 1).
- Kiyokawa, K., Billinghurst, M., Hayes, S., Gupta, A., Sannohe, Y., & Kato, H. (2002). Communication behaviors of co-located users in collaborative AR interfaces. *Proceedings. International Symposium on Mixed and Augmented Reality*, 139–148. <https://doi.org/10.1109/ISMAR.2002.1115083>
- Long, P. D., & Ehrmann, S. C. (2005). The future of the learning space: Breaking out of the box. *EDUCAUSE review*, 40(4), 42–58.
- Lu, J., & Churchill, D. (2013). Creating personal learning environments to enhance learning engagement. *2013 IEEE 63rd Annual Conference International Council for Education Media (ICEM)*, 1–8. <https://doi.org/10.1109/CICEM.2013.6820194>
- McInnerney, J. M., & Roberts, T. S. (2004). Online learning: Social interaction and the creation of a sense of community. *Journal of Educational Technology & Society*, 7(3), 73–81.
- Milgram, P., Takemura, H., Utsumi, A., & Kishino, F. (1995). Augmented reality: A class of displays on the reality-virtuality continuum. *Telemanipulator and Telepresence Technologies*, 2351, 282–292. <https://doi.org/10.1117/12.197321>
- Nørgård, R. T., & Hilli, C. (2022). Hyper-Hybrid Learning Spaces in Higher Education. In *Hybrid Learning Spaces* (pp. 25–41).

- Nuere, S., Suz, A. A., & Álvarez, L. d. M. (2021). Reflections on the adaptation to the COVID-19 pandemic in higher education. In *Remote Learning in Times of Pandemic*. Routledge.
- Özgen, D. S., Afacan, Y., & Sürer, E. (2021). Usability of virtual reality for basic design education: A comparative study with paper-based design. *International Journal of Technology and Design Education*, 31(2), 357–377.
<https://doi.org/10.1007/s10798-019-09554-0>
- Paechter, M., & Maier, B. (2010). Online or face-to-face? Students' experiences and preferences in e-learning. *The Internet and Higher Education*, 13(4), 292–297.
<https://doi.org/10.1016/j.iheduc.2010.09.004>
- Penn, M., & Ramnarain, U. (2023). A Systematic Review of Pedagogy Related to Mixed Reality in K-12 Education. *Mixed Reality for Education*, 85–108.
- Petchamé, J., Iriondo, I., Villegas, E., Riu, D., & Fonseca, D. (2021). Comparing Face-to-Face, Emergency Remote Teaching and Smart Classroom: A Qualitative Exploratory Research Based on Students' Experience during the COVID-19 Pandemic. *Sustainability*, 13(12), 6625. <https://doi.org/10.3390/su13126625>
- Petrucchio, C. (2021). Smartphones in and Outside Classroom. In *Teaching and Mobile Learning*. CRC Press.
- Popov, V., Brinkman, D., Biemans, H. J., Mulder, M., Kuznetsov, A., & Noroozi, O. (2012). Multicultural student group work in higher education. *International Journal of Intercultural Relations*, 36(2), 302–317. <https://doi.org/10.1016/j.ijintrel.2011.09.004>
- Reinius, H., Korhonen, T., & Hakkarainen, K. (2021). The design of learning spaces matters: Perceived impact of the deskless school on learning and teaching. *Learning Environments Research*, 24(3), 339–354. <https://doi.org/10.1007/s10984-020-09345-8>
- Richards, T. (2023). Reimagining the classroom : Creating new learning spaces and connecting with the world /.
- Rivoltella, P. C. (2008). Introducing Multimedia in the Classroom. In *Empowerment Through Media Education: An Intercultural Dialogue* (pp. 201–210). The International Clearinghouse on Children, Youth; Media.
- Rodden, T. (1991). A survey of CSCW systems. *Interacting with Computers*, 3(3), 319–353.
[https://doi.org/10.1016/0953-5438\(91\)90020-3](https://doi.org/10.1016/0953-5438(91)90020-3)
- Schmalstieg, D. (2016). *Augmented reality : Principles and practice* /. Addison-Wesley,
- Shank, P. (2005). The Value of Multimedia in Learning. *ADOBE DESIGN CENTER*, (501), 2–12.
- Smith, K. A., Johnson, D. W., & Johnson, R. T. (1992). Cooperative Learning and Positive Change in Higher Education. In *Collaborative Learning: A Sourcebook for Higher Education* (pp. 46–56). National Center on Postsecondary Teaching, Learning; Assessment, University Park, PA.
- Thai, N. T. T., De Wever, B., & Valcke, M. (2020). Face-to-face, blended, flipped, or online learning environment? Impact on learning performance and student cognitions. *Journal of Computer Assisted Learning*, 36(3), 397–411.
<https://doi.org/10.1111/jcal.12423>
- Van Wyk, J., & Haffejee, F. (2017). Benefits of Group Learning As a Collaborative Strategy in a Diverse Higher Education Context. *International Journal of Educational Sciences*, 18(1-3), 158–163. <https://doi.org/10.1080/09751122.2017.1305745>

- Walas-Trębacz, J., Krzyżak, J., Herdan, A., Setyohadi, D. B., Jeyanathan, J. S., & Nair, A. (2025). Social interactions and attitudes toward remote learning vs perceptions of its effectiveness in the view of Education 4.0. *The TQM Journal*, ahead-of-print(ahead-of-print). <https://doi.org/10.1108/TQM-09-2024-0335>
- Weiser, M. (1991). The Computer for the 21st Century. *Scientific American*, 265(3), 94–104. <https://doi.org/10.1038/scientificamerican0991-94>